Faculty of Engineering & Technology
Electrical & Computer Engineering Department

Digital Signal Processing

ENCS4310

Assignment#1

Correlation

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Task 1:

❖ Relation Between Convolution and Correlation

This task discusses the mathematical relation between convolution and correlation, highlights the main differences between them, and presents their typical applications in digital signal processing.

Correlation and convolution are two important operations in digital signal processing. Correlation is mainly used to measure the similarity between two signals, while convolution is used to determine the output of a system for a given input signal. Although the two operations are mathematically related, they are used for different purposes. In this task, the relation between convolution and correlation is explained, followed by the main differences and their common applications.

❖ Mathematical Relation:

Correlation:

Correlation measures the similarity between two signals as one signal is shifted relative to the other. The cross-correlation between two signals x[n] and y[n] is defined as:

$$R_{x,y}(k) = \sum x[n]y[n+k]$$

Where, k represents the shift (or delay).

Convolution:

Convolution is used to determine the output of a system when an input signal is applied. It is defined as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Where,

$x[n]: input\ signal$

$h[n]: response\ system$

$y[n]: output\ signal$

The main relation between convolution and correlation is that correlation can be expressed as convolution with a time-reversed version of one of the signals.

$$R_{x,y}(k) = \sum_{n=-\infty}^{\infty} x[n]y[n+k]$$

This is equivalent to convolving x[n] with the time-reversed signal y[−n]:

$$R_{x,y}(k) = x[n] * y[-n]$$

Therefore, correlation measures similarity between signals, while convolution describes how a system modifies an input signal.

- **Key Differences:**

Although correlation and convolution are mathematically related, they differ in their purpose, interpretation, and usage in signal processing.

The main difference between correlation and convolution lies in their purpose and interpretation. Correlation is primarily used to measure the similarity between two signals, making it useful in applications such as signal detection and pattern recognition. In contrast, convolution is used to determine the output of a system when an input signal is applied, which makes it essential in system analysis and filtering. Another important difference is that convolution explicitly involves time-reversing one signal before shifting and multiplying, while correlation is mainly interpreted as a similarity measure between signals. Mathematically, correlation can also be represented using a time-reversed version of one signal. Therefore, correlation focuses on measuring similarity between signals, whereas convolution describes how signals are transformed by systems.

- **Typical Applications:**

Both correlation and convolution are widely used in digital signal processing, but each is applied in different types of problems depending on its purpose.

Correlation is mainly used in applications that involve measuring similarity between signals. It is commonly used in signal detection, where the goal is to determine whether a known signal is present within another signal. It is also used in pattern recognition, image matching, and time delay estimation. For example, in radar systems, correlation is used to detect reflected signals and determine their delay.

Convolution is used in applications related to system analysis and signal processing. It is essential in determining the output of linear time-invariant (LTI) systems. Convolution is widely used in filtering, where signals are processed to remove noise or extract useful information. It is also used in image processing, communication systems, and control systems.

In general, correlation is preferred when comparing signals, while convolution is used when analyzing how systems affect signals.

- **Similarity Detection Problem**

Similarity detection using correlation refers to the process of determining how similar one signal is to another. In many real-world applications, we often have a known signal and want to check whether it exists within another signal, or which signal is closest to it in shape. Correlation provides a numerical way to measure this similarity.

In simple terms, correlation works by comparing two signals element by element. If the signals have similar patterns or shapes, the correlation value will be high. If they are different, the value will be smaller, and if they are opposite in behavior, the value may become negative. Therefore, correlation helps us decide which signal is most similar to a reference signal.

Standard Correlation

To measure similarity between signals, we first use standard correlation. This method compares two signals by multiplying their corresponding elements and summing the results into a single value. The higher this value, the more similar the signals are considered to be.

For time-limited signals, the correlation at zero shift is given by:

$$R_{x,y}(0) = \sum_{n=0}^{N-1} x[n]y[n]$$

This means that each element of the first signal is multiplied by the corresponding element of the second signal, and then all the products are added together. The result is a single number that represents how similar the two signals are.

Given the following signals:

$x = [1\ 3 - 2\ 4]$

$y = [2\ 3 - 1\ 3]$

$z = [2\ \ -1\ \ 4 - 2]$

$w = [100 - 1\ 4\ \ -2]$

$$R_{x,y} = 1(2) + 3(3) + (-2)(-1) + 4(3) = 25$$

$$R_{x,z} = 1(2) + 3(-1) + (-2)(4) + 4(-2) = -17$$

$$R_{x,w} = 1(100) + 3(-1) + (-2)(4) + 4(-2) = 81$$

From the above results, we notice that $R_{x,w} = 81$, which is greater than $R_{x,y} = 25$. This suggests that signal w is more similar to x than y. However, this conclusion is incorrect.

The reason is that the value 100 in signal w is very large compared to the other values. This large value dominates the calculation and artificially increases the correlation result. Therefore, even though w is not truly more similar to x, the standard correlation gives a misleading result.

This shows that standard correlation is strongly affected by the amplitude (magnitude) of the signal values, and it may not always reflect the true similarity between signals.

Normalized Correlation

As shown earlier, standard correlation can give misleading results when one of the signals contains very large values. To solve this problem, normalized correlation is used.

Normalized correlation adjusts the result by taking into account the overall energy (or magnitude) of each signal. This allows us to compare signals based on their shape rather than their absolute values.

$$\sigma_{x,y} = \frac{\sum x[n]y[n]}{\sqrt{\sum x^2[n]} \, . \sqrt{\sum y^2[n]}}$$

This formula divides the correlation value by the product of the magnitudes of both signals. As a result, the final value is normalized and typically lies between -1 and 1.

To demonstrate the effectiveness of normalized correlation in measuring signal similarity, we apply the formula to the given signals step by step.

$$\sum x[n]y[n] = 1(2) + 3(3) + (-2)(-1) + 4(3) = 25$$

$$\sum x^2[n] = 1^2 + 3^2 + (-2)^2 + 4^2 = 30$$

$$\sum y^2[n] = 2^2 + 3^2 + (-1)^2 + 3^2 = 23$$

$$\sigma_{x,y} = \frac{25}{\sqrt{30}.\sqrt{23}} = 0.952$$

$$\sum x[n]z[n] = 1(2) + 3(-1) + (-2)(4) + 4(-2) = -17$$

$$\sum z^2[n] = 2^2 + (-1)^2 + 4^2 + (-2)^2 = 25$$

$$\sigma_{x,z} = \frac{-17}{\sqrt{30}.\sqrt{25}} = -0.621$$

$$\sum x[n]w[n] = 1(100) + 3(-1) + (-2)(4) + 4(-2) = 81$$

$$\sum w^2[n] = 100^2 + (-1)^2 + 4^2 + (-2)^2 = 10021$$

$$\sigma_{x,w} = \frac{81}{\sqrt{30}.\sqrt{10021}} = 0.148$$

The normalized correlation results are approximately 0.952 for x and y, −0.621 for x and z, and 0.148 for x and w.

These results provide a much more accurate comparison. The signal y has the highest normalized correlation with x, which correctly indicates that it is the most similar signal. In contrast, the signal w, which previously had the highest standard correlation, now shows a much lower value.

This confirms that normalized correlation eliminates the misleading effect of large values and focuses on the actual similarity in signal shape.

Normalization improves detection because it removes the effect of large amplitudes and ensures that the comparison is based on the shape of the signals rather than their magnitude. This leads to more reliable similarity measurements.

Normalized correlation should be used when comparing signals that may have different amplitudes or energy levels. It is especially useful in signal detection, pattern recognition, and template matching applications.

Correlation is an effective tool for measuring similarity between signals, but it can be affected by amplitude. Normalized correlation provides a more reliable approach by focusing on the signal shape rather than magnitude.

Task 2:

- **Detecting the Presence of s1 Using Correlation**

In many signal processing applications, it is important not only to compare signals, but also to determine whether a specific signal is present within a more complex signal, and how strongly it appears. This process is known as signal detection.

In this task, the signal s1 is treated as a reference signal, and the goal is to evaluate its presence within two composite signals, X and Y. These signals are formed by combining multiple sinusoidal components with different amplitudes and frequencies. Therefore, the challenge is to accurately measure how much of s1 exists within each signal.

To achieve this, both standard correlation and normalized correlation are used and compared. While standard correlation may produce high values due to large signal amplitudes, normalized correlation provides a more reliable measure by focusing on the similarity in signal shape rather than magnitude. This comparison helps determine which method better reflects the true strength of the signal's presence.

- **Signal Generation**

To analyze the presence of the signal s1, the signals must first be generated. The time vector is defined over one second with a sampling interval of 0.01 seconds. Three sinusoidal signals s1, s2, and s3 are then generated using cosine functions with different frequencies.

The signals are defined as follows:

$$s1 = \cos(2\pi.1.t)$$

$$s2 = \cos(2\pi.4.t)$$

$$s3 = \cos(2\pi.10.t)$$

Using these signals, two composite signals are constructed:

$$X = 2s1 + 4s2 + s3$$

$$Y = s1 + s2$$

These signals are used to evaluate how strongly the signal s1 is present within each composite signal.

- ❖ **Plotting the Generated Signals**

To better understand the structure of the generated signals, all signals are plotted over a time interval of one second. Plotting helps visualize how each sinusoidal component behaves and how the composite signals X and Y are formed from these components.

Result:

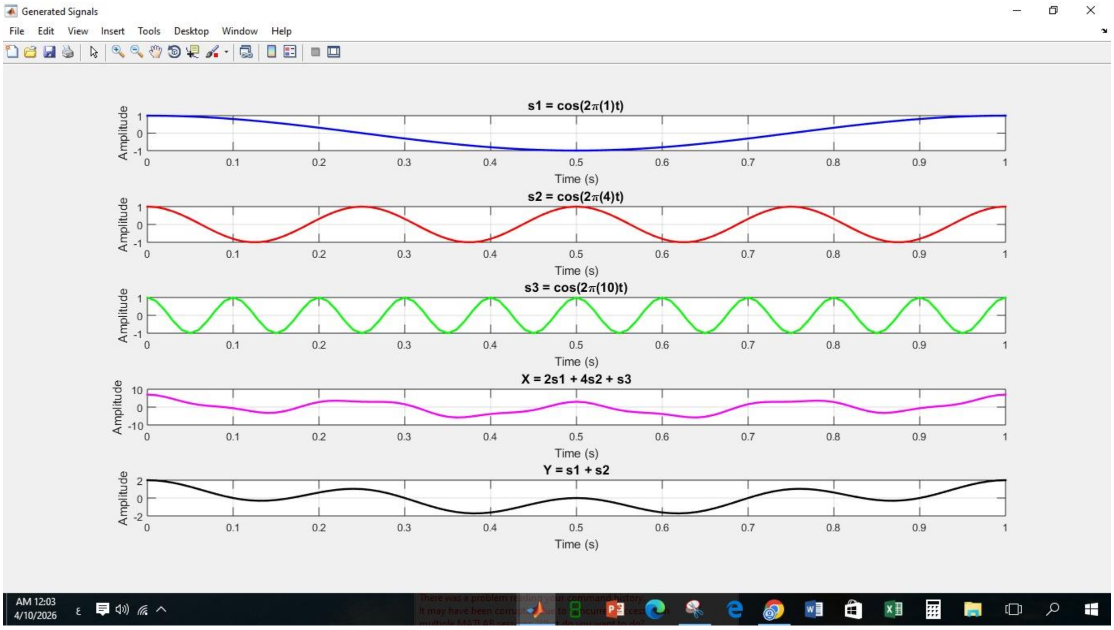

Figure 1: Generated signals over 1 second, including s1, s2, s3, and the composite signals X and Y.

From the plots, it can be observed that each signal has a different frequency, where s1 is the slowest signal and s3 is the fastest. The composite signal X appears more complex due to the presence of three components with different amplitudes, especially the stronger contribution of s2. On the other hand, Y is simpler since it contains only two components. These visual differences suggest that identifying the presence of s1 may not be straightforward using standard correlation alone.

❖ Correlation Between X and s1

To evaluate how strongly the signal s1 is present within the composite signal X, both standard correlation and normalized correlation are computed. These two methods are compared to determine which one better reflects the true presence of s1.

Standard Correlation:

To evaluate the presence of the signal s1 within the composite signal X, standard correlation is first computed using the following equation:

$$R(X, s1) = \sum X[n]s1[n]$$

```
% Standard Correlation
R_X_s1 = sum(X .* s1);
fprintf('\n--- Standard Correlation ---\n');
fprintf('R(X, s1) = %.4f\n', R_X_s1);
```

Figure 2: compute standard correlation R(X,s1)

```
--- Standard Correlation ---
R(X, s1) = 107.0000
```

Figure 3: Result

Using MATLAB, the computed value is:

$$R(X, s1) = 107.0000$$

This relatively high value suggests that the signal s1 is present within X. However, this result is influenced by the overall amplitude of the signal X. Since X contains components with larger amplitudes, especially the term 4s2, the standard correlation value becomes large even if the actual similarity is not very strong.

Normalized Correlation:

To overcome the limitation of standard correlation, normalized correlation is used. It is defined as:

$$\sigma_{x,s1} = \frac{\sum X[n]s1[n]}{\sqrt{\sum X^2[n]} . \sqrt{\sum s1^2[n]}}$$

```
% Normalized Correlation
fprintf('\n--- Normalized Correlation ---\n');
R_X_s1 = sum(X .* s1);
norm_X  = sqrt(sum(X.^2));
norm_s1 = sqrt(sum(s1.^2));
sigma_X_s1 = R_X_s1 / (norm_X * norm_s1);
fprintf('sigma(X, s1) = %.4f\n', sigma_X_s1);
```

Figure 4: compute normalized correlation σ(X,s1)

```
--- Normalized Correlation ---
sigma(X, s1) = 0.4520
```

Figure 5:result

The computed normalized correlation value is:

$$\sigma(X, s1) = 0.4520$$

Unlike standard correlation, this value is independent of signal amplitude and reflects the actual similarity between the signals. The value 0.4520 indicates that although s1 is present in X, the similarity is moderate rather than strong. This is because X also contains other components with different frequencies that affect its overall shape.

This demonstrates that normalized correlation provides a more accurate measure of signal similarity, as it eliminates the misleading effect of large amplitudes present in the signal.

- ❖ **Correlation Between Y and s1**

To further evaluate the presence of the signal s1, the correlation is computed between Y and s1 using standard correlation:

Standard Correlation:

$$R(Y, s1) = \sum Y[n]s1[n]$$

```
% Standard Correlation
R_Y_s1 = sum(Y .* s1);
fprintf('\n--- Standard Correlation ---\n');
fprintf('R(Y, s1) = %.4f\n', R_Y_s1);
```

Figure 6: compute standard correlation R(Y,s1)

```
--- Standard Correlation ---
R(Y, s1) = 52.0000
```

Figure 7: result

Using MATLAB, the computed value is:

$$R(Y, s1) = 52.0000$$

This value is smaller than the standard correlation obtained for X. However, this does not necessarily mean that s1 is less present in Y. The standard correlation is influenced by the overall amplitude of the signal, and since X contains higher amplitude components, it produces a larger correlation value.

Normalized Correlation:

To overcome the limitation of standard correlation, normalized correlation is used. It is defined as:

$$\sigma_{Y,s1} = \frac{\sum Y[n]s1[n]}{\sqrt{\sum Y^2[n]} \cdot \sqrt{\sum s1^2[n]}}$$

```
% Normalized Correlation
fprintf('\n--- Normalized Correlation ---\n');
R_Y_s1 = sum(Y .* s1);
norm_Y  = sqrt(sum(Y.^2));
sigma_Y_s1 = R_Y_s1 / (norm_Y * norm_s1);
fprintf('sigma(Y, s1) = %.4f\n', sigma_Y_s1);
```

Figure 8: compute normalized correlation σ(Y,s1)

```
--- Normalized Correlation ---
sigma(Y, s1) = 0.7140
```

Figure 9: result

The computed normalized correlation value is:

$$\sigma(Y, s1) = 0.7140$$

Unlike standard correlation, the normalized correlation for Y is higher than that of X. This indicates that the signal s1 is actually more clearly present in Y when the effect of amplitude is removed.

This result confirms that normalized correlation provides a more accurate measure of similarity, as it focuses on the shape of the signal rather than its magnitude.

❖ Comparison Between Standard and Normalized Correlation

Standard Correlation Plotting:

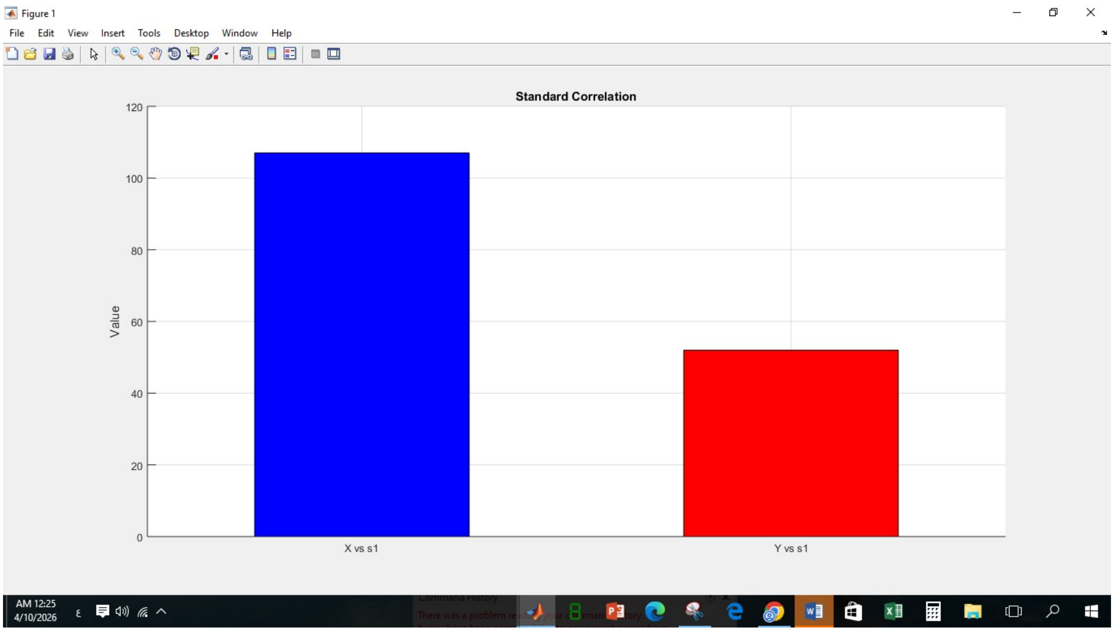

Figure 10: Standard correlation values between the signals X and Y with respect to the reference signal s1

It can be observed that the value of R(X,s1) is significantly higher than R(Y,s1). This may suggest that the signal s1 is more strongly present in X.

However, this conclusion is misleading. The signal X contains components with larger amplitudes, especially the term 4s2, which increases the overall magnitude of the signal. As a result, the standard correlation value becomes larger due to amplitude rather than actual similarity.

Normalized Correlation Plotting:

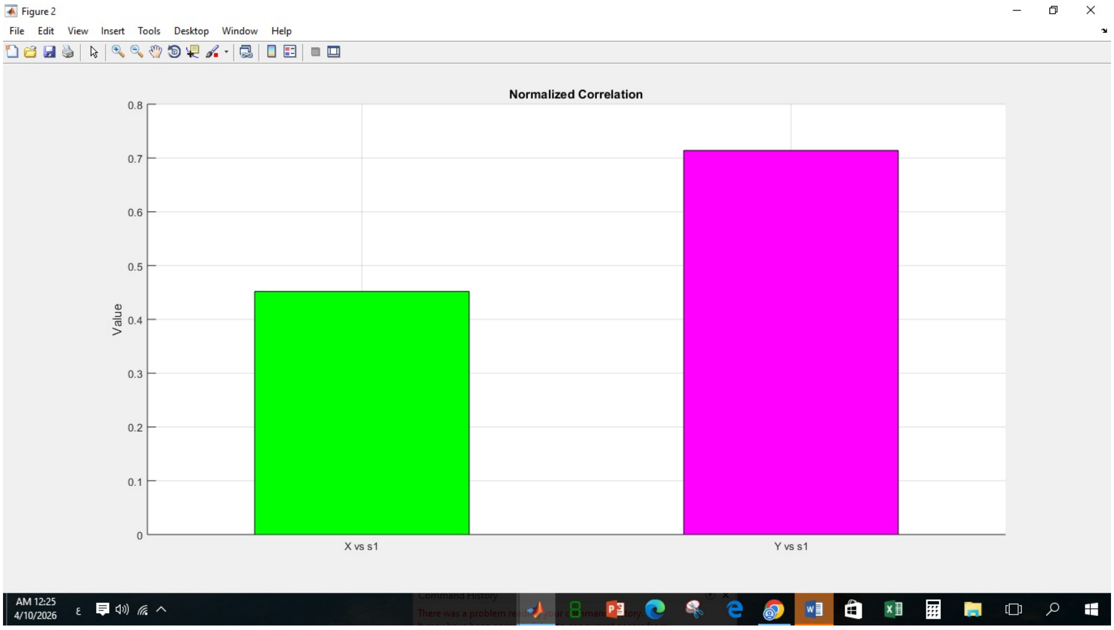

Figure 11: Normalized correlation values between the signals X and Y with respect to the reference signal s1

Unlike standard correlation, the value of σ(Y,s1) is higher than σ(X,s1). This indicates that the signal s1 is actually more clearly present in Y when the effect of amplitude is removed.

Normalized correlation provides a fair comparison by eliminating the influence of signal magnitude and focusing only on the similarity in signal shape. Therefore, it gives a more accurate representation of the true presence of s1.

From the comparison, it is clear that standard correlation can give misleading results when signals have different amplitudes. In contrast, normalized correlation provides a more reliable measure of similarity and is better suited for detecting the presence of a signal.

- **Effect of Signal Amplitude on Standard Correlation**

Standard correlation depends on amplitude because it directly sums the product of signal values. Therefore, signals with larger amplitudes contribute more to the result, even if the actual similarity is not strong.

❖ Preferred Method for Signal Detection and Pattern Recognition

Based on the results obtained from the analysis, normalized correlation is the preferred method for both signal detection and pattern recognition.

In signal detection, the goal is to determine whether a specific signal is present within another signal. Normalized correlation is more suitable because it eliminates the effect of signal amplitude and focuses on the similarity in signal shape. This allows for a more accurate detection of the signal presence.

Similarly, in pattern recognition, the objective is to compare patterns based on their structure rather than their magnitude. Therefore, normalized correlation is more effective, as it provides a reliable measure of similarity regardless of the signal strength.

In contrast, standard correlation may give misleading results due to its dependence on amplitude, making it less reliable for these applications.

Appendix:

Code:

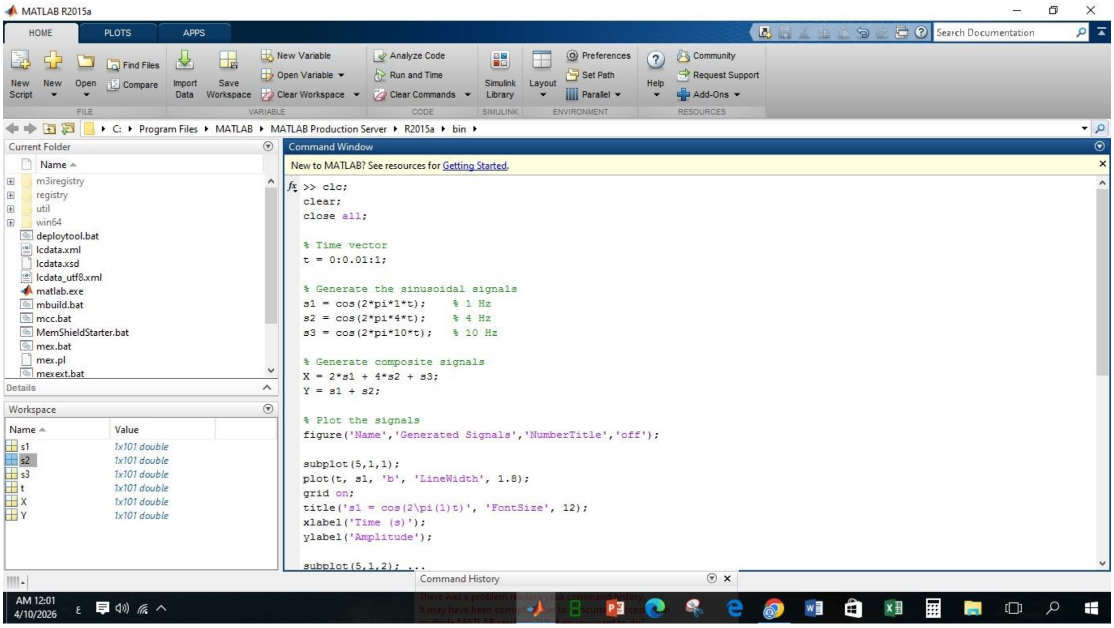

Figure 12: code1

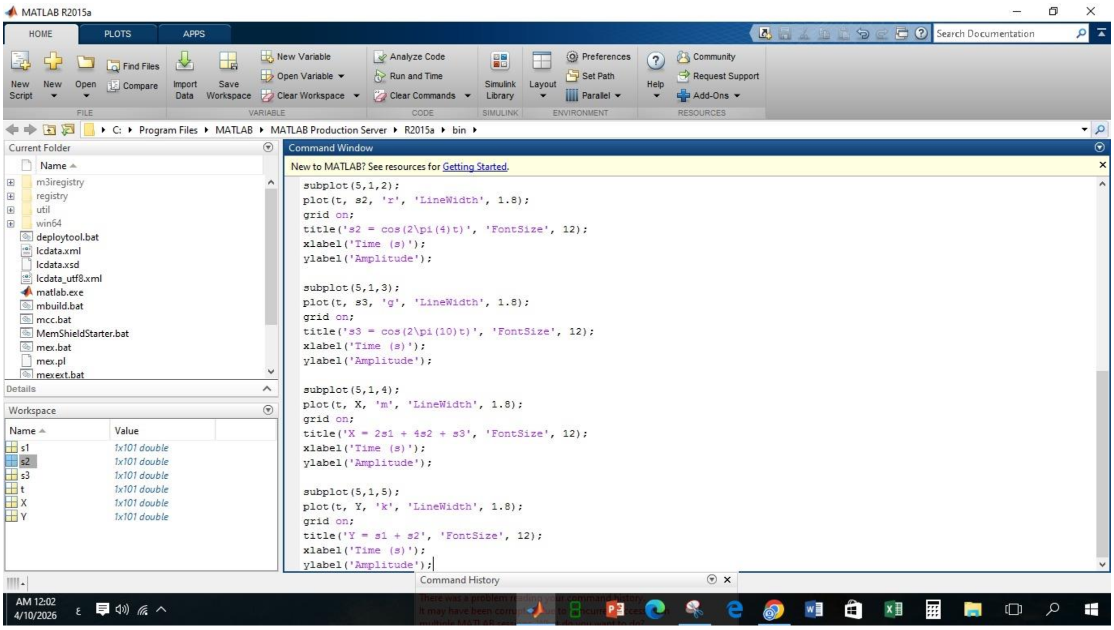

Figure 13: code2

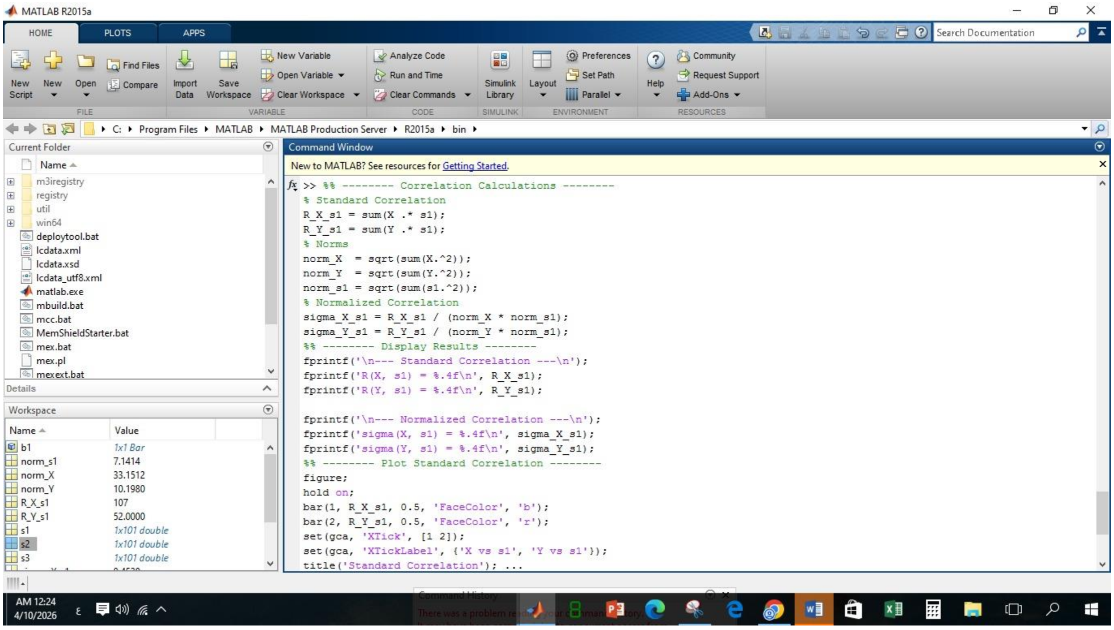

Figure 14: code3

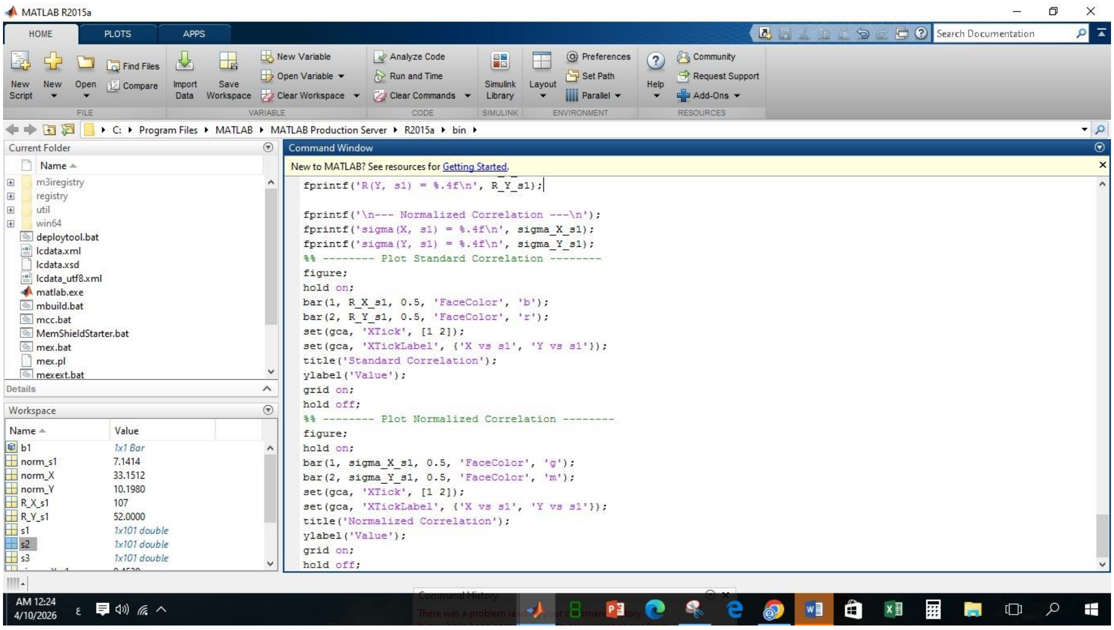

Figure 15: code4